Building Reusable Analytical Systems: Testing Data Continuity Across ML and NLP Workflows

Understanding Reusable Continuity

Across ML and NLP Systems

Siddhartha Jain

Siddharthajain956@gmail.com

2025-2026

Contents

1. Project Overview

This project focuses on understanding whether the Machine Learning (ML) and Natural Language Processing (NLP) workflows that I had previously developed can function as reusable analytical systems instead of remaining one-time project structures. The objective was to explore how future market data and future text data can enter the same workflows without requiring the entire analytical process to be rebuilt each time new information becomes available.

While working on the earlier ML and NLP projects, I realised that most analytical workflows are designed around a specific dataset and often become static after completion. However, financial and textual data continuously evolve over time, making long-term continuity an important consideration. This project therefore focuses on understanding whether the existing workflows can continue operating when new data is introduced while preserving their analytical structure and continuity.

Since the ML and NLP systems follow different architectural approaches, the project also examines how both workflows behave operationally when future data enters the system. Overall, the project aims to understand how reusable analytical systems can be created, maintained, and expanded over time using the structures that already exist within the ML and NLP workflows.

2. Problem Statement

While working on the Machine Learning (ML) and Natural Language Processing (NLP) projects, I realised that most analytical workflows are built around a fixed dataset and often remain unchanged once the analysis is complete. Although these projects successfully answered their original analytical questions, they raised another important question: what happens when new market data, news data, or earnings data becomes available in the future? If new information continuously enters the system, can the existing workflows continue operating effectively, or will significant parts of the process need to be rebuilt each time?

This question became more important because many analytical and financial modelling workflows are designed primarily to solve a problem at a specific point in time. While these workflows may perform well initially, maintaining them over longer periods often requires additional manual effort. As new data becomes available, datasets may need to be refreshed, calculations may need to be updated, and analytical outputs may need to be regenerated. This creates a challenge for long-term continuity because the workflow can gradually become dependent on repeated manual intervention.

While reflecting on the ML and NLP projects, I started questioning whether the existing structures could behave differently. Instead of focusing on creating new models or new analytical signals, I became interested in understanding whether the workflows themselves could be reused. If future data was appended to the existing datasets, would the downstream views, analysis layers, prediction structures, and signal workflows continue functioning correctly, or would the overall process become unstable and require rebuilding?

The problem became particularly important because the two projects were built using very different architectures. The ML workflow operates through a highly connected structure within BigQuery SQL and BigQuery ML, where models, views, and downstream analysis remain closely linked. The NLP workflow follows a more modular structure where external Python processing is required before the extracted outputs are integrated back into SQL for further analysis. Since both systems operate differently, understanding their ability to support future data continuity became an important challenge.

Therefore, the central problem of this project is to determine whether the existing ML and NLP workflows can function as reusable analytical systems rather than remaining static analytical projects. More specifically, the project seeks to understand whether future market data and future text data can be integrated into the existing architecture while preserving operational continuity, structural stability, and analytical usefulness over time.

3. Dataset and Project Setup

Before testing whether the existing ML and NLP workflows could function as reusable analytical systems, it was important to understand the datasets, tools, and operational structure on which both projects were built. Since this project focuses on reusing and testing the existing analytical architecture rather than creating a new one, the same datasets and environments used in the earlier projects became the foundation for all four major questions of this project.

Table 1: Dataset Summary

Dataset	Purpose
Market Return Dataset	Used as the primary structured dataset for the ML workflow and as a market reference dataset in several NLP analyses.
News Dataset	Used for sentiment, entity, phrase, and theme-related NLP signal extraction and analysis.
Earnings Dataset	Used for earnings sentiment extraction and earnings-related NLP analysis.

The Machine Learning workflow was built using historical market data consisting of Nifty index returns and return data for five selected stocks. This dataset served as the primary input for the ML Project. The workflow was implemented within BigQuery SQL and BigQuery ML, where models, views, predictions, and analysis layers were connected through a dependency-based structure.

The Natural Language Processing workflow was built using the market-return dataset together with the news and earnings datasets. The news dataset was used to generate sentiment, entity, phrase, and theme-related signals, while the earnings dataset was used for earnings sentiment analysis. Unlike the ML workflow, the NLP workflow relied on external Python processing before the extracted outputs were integrated into BigQuery SQL for further analysis.

The project environment consisted primarily of BigQuery SQL, BigQuery ML, Python, Jupyter Notebook, and Microsoft Excel. BigQuery SQL was used for data storage, views, and downstream analysis. BigQuery ML supported machine learning model development and prediction-related analysis. Python and Jupyter Notebook were used for data extraction, preprocessing, and NLP signal generation, while Microsoft Excel supported preprocessing and return calculations where required.

Another important aspect of the setup was the organizational structure of the workflows. Both projects followed a Scrum-style approach in which larger analytical objectives were divided into major questions, sub-questions, and smaller operational tasks. Views and checkpoints helped organize these activities into manageable stages while maintaining continuity across the overall workflow.

Overall, the datasets, tools, and organizational structure formed the foundation of the current project. Every stage of the project, from architecture understanding and schema validation to ML and NLP continuity testing, was built upon these existing components.

4. Analytical Path / Project Method

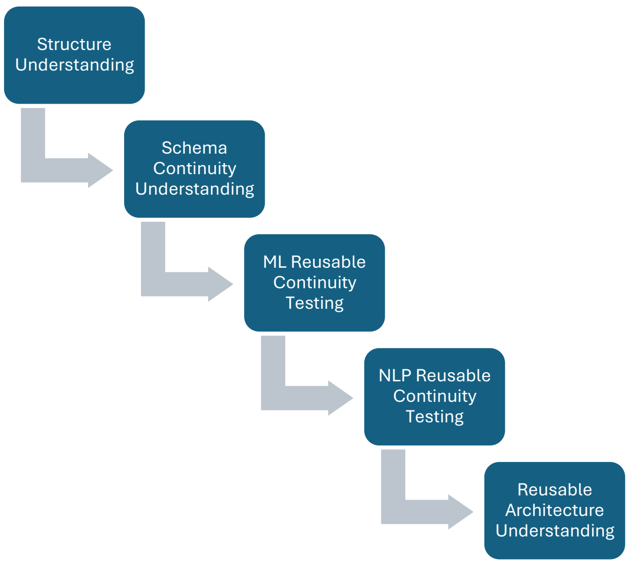
After understanding the datasets, tools, and overall project setup, the next step was to determine how the project should progress. Since the objective was to test the reusable continuity of the existing ML and NLP workflows, it was important to follow a structured approach rather than directly introducing new data into the systems. Without understanding the architecture first, it would be difficult to identify the source of any operational issues that might emerge later in the workflow.

For this reason, the project follows a staged progression where each stage creates the foundation for the next. The first stage focuses on understanding the existing structure of the ML and NLP workflows. The second stage focuses on understanding schema continuity and the conditions required for future data to enter the system safely. Once these foundations are established, the project moves into implementation by testing reusable continuity within the ML workflow and then within the NLP workflow.

This sequence was designed to ensure that the implementation stages were supported by a clear understanding of both the architecture and the operational requirements of the system. Rather than treating reusable continuity as a purely technical exercise, the project approaches it as a process of understanding how the existing analytical architecture behaves when future data is introduced.

Overall, the project follows a simple progression: understanding the existing structure, understanding schema continuity, testing reusable continuity in the ML workflow, and finally testing reusable continuity in the NLP workflow. Together, these stages provide a structured path for evaluating whether the existing analytical architecture can function as a reusable system over time.

Figure 1. Project Continuity Testing Framework



5. Major Question 1 — Understanding the Existing Reusable Structure of the ML and NLP Systems

5.1 The Question

The first major question of this project focuses on understanding the existing structure of the ML and NLP workflows before attempting to test their reusable continuity. Since both projects had already been developed and operationalized, it was important to understand how the different datasets, views, models, signals, and analytical layers were connected within each workflow. Before determining whether a system can continue functioning when future data is introduced, I first needed to understand how that system currently operates. This question therefore focuses on identifying the structure, dependencies, operational layers, and continuity mechanisms that already exist within both projects.

5.2 Why This Stage Was Needed

It would have been possible to immediately append new data into the ML and NLP workflows and observe the results. However, doing so without first understanding the architecture would make it difficult to identify the source of any problems that might emerge. If an error appeared after introducing new data, the issue could originate from the raw datasets, the schema structure, the views, the models, the NLP signals, or even from an incorrect understanding of how the workflow itself operates.

This stage was therefore necessary because reusable continuity depends on more than simply adding new data into a system. It requires an understanding of how different components interact with one another and where the major operational dependencies exist. Before testing whether the workflows could survive future data continuity, I first needed to understand the architecture that was expected to support that continuity.

5.3 Understanding the Existing Architecture

Understanding the Existing Project Structure

The first step was reviewing the overall structure of both projects. While studying the ML workflow, I observed that it was built as a highly connected architecture where the raw input dataset, machine learning models, views, predictions, and downstream analytical layers were linked together through a common structure. Most of the workflow operated within BigQuery SQL and BigQuery ML, creating a centralized environment where different components remained closely connected.

The NLP workflow followed a different structure. Instead of one centralized environment, the workflow combined external Python processing with downstream SQL analysis.

Signals such as sentiment, entities, phrases, and earnings sentiment were generated separately before being integrated into BigQuery SQL for further analysis. Because of this separation between extraction and analysis, the NLP workflow behaved as a more modular system compared to the ML workflow.

Reusable and Manual Layers

After understanding the structure, I examined which parts of the workflows were already reusable and which parts still depended on manual intervention. Initially, many components appeared manual because they required active operational involvement. However, as I reviewed the architecture more carefully, I realised that several parts of both workflows were already functioning in a reusable manner.

The ML workflow contained a large reusable layer because the views, models, predictions, and analytical outputs remained connected through the existing structure. Once the data entered the workflow correctly, much of the downstream continuity could operate through the same architecture. The NLP workflow also contained reusable elements, particularly within the extraction processes, but it continued to rely more heavily on external processing and signal regeneration. Some analytical activities, especially theme interpretation and phrase grouping, remained partially manual because they required analytical judgment rather than purely technical execution.

Operational Flow and Continuity

The next step was understanding how information moved through both systems. In the ML workflow, the data flow followed a relatively centralized path where the raw dataset acted as the foundation for models, predictions, and downstream analytical layers. Since most stages existed within the same environment, the operational continuity depended heavily on maintaining the integrity of the underlying structure.

The NLP workflow followed a different path. Raw news and earnings data were first processed through Python-based NLP functions before the extracted outputs were integrated into BigQuery SQL. Because of this separation, the continuity of the NLP workflow depended on maintaining stable connections between the extraction stage and the analytical stage. Understanding these operational flows helped clarify where continuity existed and where disruptions were most likely to occur if future data entered the workflow incorrectly.

Key Architectural Understanding

The most important understanding from this stage was that the ML and NLP workflows survive continuity in different ways. The ML workflow relies primarily on centralized dependency-connected continuity, where most components remain linked within the same environment. The NLP workflow relies more heavily on modular continuity, where individual signals are generated separately and later integrated into the broader analytical structure.

I also realised that continuity depends not only on the datasets themselves but also on the connection points that allow information to move through the workflow. These connection points act as the operational bridges that preserve the relationship between different stages of the architecture. Understanding these differences became important because they define how reusable continuity can realistically be supported within each system.

5.4 Key Findings

The analysis revealed that both the ML and NLP workflows already contain significant reusable components, although they achieve continuity through different architectural approaches. The ML workflow operates as a centralized dependency-connected system where models, views, predictions, and downstream analytical layers remain closely linked through a common structure. This creates strong continuity but also increases dependence on maintaining the integrity of the overall architecture.

The NLP workflow behaves as a modular signal-based system where multiple extraction processes generate independent outputs before they are integrated into SQL for further analysis. This architecture provides greater separation between components but also introduces additional operational steps that must remain compatible with one another.

Another important finding was that reusable continuity and complete automation are not necessarily the same thing. Several operational processes can remain reusable even when certain analytical tasks continue to require human judgment. This was particularly visible within the theme-analysis components of the NLP workflow.

Overall, both projects already possessed the foundations of reusable analytical systems, but the mechanisms through which continuity was maintained differed significantly between them.

5.5 Interpretation

The findings from this stage provided an important shift in understanding. Initially, I viewed the ML and NLP projects primarily as completed analytical projects designed to answer specific questions. However, after examining their structure, operational layers, and continuity mechanisms, I began to view them as systems rather than individual analyses.

The ML workflow demonstrated how reusable continuity can emerge through a highly connected architecture where downstream components remain synchronized through shared dependencies. The NLP workflow demonstrated a different form of continuity where reusable extraction processes and modular signal generation support the overall analytical structure. Although both systems behave differently, they are ultimately attempting to achieve the same objective: allowing future data to move through an existing analytical framework without requiring the entire workflow to be rebuilt.

This understanding became the foundation for the remainder of the project. Before testing schema continuity, ingestion frameworks, or appended-data workflows, it was first necessary to understand how the architecture itself behaved. Only after understanding these operational foundations could the reusable continuity of the systems be evaluated in a meaningful way.

6. Major Question 2 — Understanding Schema Continuity and Building Reusable Ingestion Frameworks

6.1 The Question

After understanding the existing structure of the ML and NLP workflows, the next major question focused on understanding the conditions required for these systems to remain reusable when future data is introduced. While the architecture of a workflow may remain unchanged, its continuity can still fail if new data does not enter the system in a form that the workflow expects. This question therefore focused on understanding schema continuity, validation requirements, and reusable ingestion frameworks to determine how future data could be integrated into the existing architecture without disrupting the operational continuity of the system.

6.2 Why This Stage Was Needed

The first major question established how the ML and NLP workflows were structured and how continuity was maintained within each system. However, understanding the architecture alone was not enough to determine whether the workflows could remain reusable over time. Even a well-designed workflow can become unreliable if future data does not match the structural requirements on which it was originally built.

This stage became necessary because future market data, news data, and earnings data are naturally expected to change. The values, patterns, topics, and signals may all evolve over time. However, if the workflow is expected to survive these changes, it must be able to accept new data without breaking the existing analytical structure. Before testing reusable continuity in practice, it was therefore important to understand the structural conditions that would allow future data to enter the workflow safely and consistently.

6.3 Analysis and Understanding

Schema Continuity Understanding

The first step was understanding what schema continuity actually means within the context of this project. Initially, it seemed reasonable to assume that reusable continuity depended mainly on the future data itself. However, while reviewing both workflows, I gradually realised that the architecture was far more dependent on structure than on content.

The workflows had been built around specific column structures, naming conventions, data types, and relationships between datasets. While future market behaviour may change and future news topics may evolve, the workflow only expects the incoming data to follow the structural rules on which it was originally designed. This led to an important realization: reusable continuity does not depend on preserving the same data over time. Instead, it depends on preserving the structure through which new data enters the system.

Validation and Compatibility Requirements

Once the importance of schema continuity became clear, the next step was understanding how that continuity could be protected. Both workflows contained multiple downstream dependencies that relied on consistent inputs. Views, models, analytical layers, and NLP signal-generation processes all expected data to follow specific structural requirements.

While reviewing these dependencies, I realised that validation is not simply a technical checkpoint. It acts as a mechanism that protects the workflow from structural inconsistencies before they propagate downstream. Without validation, a problem introduced at the input stage could affect multiple parts of the architecture. Validation therefore became important because it helps preserve compatibility between incoming data and the existing workflow structure.

Reusable Ingestion Framework Understanding

After understanding schema continuity and validation, the next focus was understanding how future data could repeatedly enter the system. Rather than manually redesigning the process whenever new data became available, I explored the idea of reusable ingestion frameworks. These frameworks provide a consistent process through which future data can be collected, prepared, and integrated into the workflow.

Within the ML workflow, this understanding supported the creation of reusable market-data ingestion processes. Within the NLP workflow, it highlighted the need for repeatable extraction structures for future news and earnings data. Although the workflows differ architecturally, both depend on having a reliable entry mechanism through which future data can be introduced without disturbing the existing structure.

Key Operational Understanding

The most important understanding from this stage was that reusable continuity survives through structure rather than through data itself. Future data is expected to change continuously. Market conditions will evolve, news narratives will shift, and earnings discussions will introduce new themes over time. The objective of a reusable system is therefore not to preserve the data, but to preserve the conditions under which the workflow can continue operating when new data arrives.

This understanding changed the way I viewed reusability. Rather than seeing it as an automation problem, I began to view it as a continuity problem. Schema continuity, validation processes, and ingestion frameworks all contribute to preserving the structural stability that allows the architecture to remain operational despite changes in the underlying data.

6.4 Key Findings

The analysis revealed that schema continuity is one of the most important requirements for maintaining reusable analytical systems. Both the ML and NLP workflows depend on stable structural inputs in order to preserve downstream compatibility and operational continuity.

The ML workflow was found to be particularly dependent on schema continuity because of its highly connected architecture. Structural inconsistencies introduced at the raw-input level have the potential to affect models, views, predictions, and downstream analytical layers. The NLP workflow demonstrated greater modularity, but it also remained dependent on maintaining consistency across extraction processes and signal-generation workflows.

Another important finding was that validation serves as a continuity-preservation mechanism rather than simply a quality-control activity. The analysis also showed that reusable ingestion frameworks provide the structured pathway through which future data can repeatedly enter the workflow while maintaining compatibility with the existing architecture.

Overall, schema continuity, validation, and ingestion frameworks emerged as foundational requirements for reusable continuity across both systems.

6.5 Interpretation

This stage significantly changed my understanding of what makes an analytical system reusable. At the beginning of the project, I viewed reusability mainly as the ability to introduce new data into an existing workflow. However, the analysis showed that the real challenge lies in preserving the structural conditions that allow the workflow to function correctly.

One of the most important realizations was that future data does not need to remain similar for continuity to survive. Market behaviour can change, news topics can evolve, and earnings discussions can introduce entirely new themes. The workflow can still remain operational if the incoming data continues to follow the structural rules expected by the system. In this sense, reusable continuity depends more on preserving structure than on preserving data.

This understanding established an important foundation for the remainder of the project. Before testing reusable continuity within the ML and NLP workflows, it became necessary to understand the structural requirements that make such continuity possible. Schema continuity, validation, and ingestion frameworks therefore emerged as the operational foundations upon which future continuity testing would depend.

7. Major Question 3 — Testing Reusable Continuity in the ML Workflow

7.1 The Question

After understanding the architecture of the ML workflow and identifying the structural conditions required for continuity, the next major question focused on testing whether reusable continuity could be achieved in practice. Rather than discussing reusability conceptually, this stage aimed to determine whether future market data could enter the existing ML workflow while preserving the functionality of the downstream architecture. The question therefore became whether the existing ML system could continue operating through the same models, views, predictions, and analytical layers when new data was introduced without rebuilding the workflow.

7.2 Why This Stage Was Needed

The earlier stages of the project established how the ML workflow was structured and why schema continuity was necessary for preserving operational stability. However, understanding these principles alone was not sufficient to demonstrate that reusable continuity actually exists. The concept needed to be tested within the workflow itself.

The ML workflow became the first implementation stage because it operates through a highly connected architecture where multiple downstream components depend on the same underlying data structure. If reusable continuity could survive within such an environment, it would provide practical evidence that the workflow could continue functioning beyond the original dataset on which it was built.

This stage therefore focused on introducing future market data into the workflow and observing whether the existing architecture, models, views, predictions, and analytical layers remained operational without requiring reconstruction.

7.3 Analysis and Understanding

Reusable Market Data Extraction Framework

The first step involved creating a reusable process for collecting future market data. Rather than manually rebuilding the dataset whenever new information became available, I developed a reusable extraction framework using Python and Yahoo Finance. The framework was designed around reusable parameters such as tickers, start dates, and end dates, allowing future market data to be collected through the same operational process.

This approach transformed data collection from a one-time activity into a repeatable workflow. Instead of creating a new extraction process for every update, the same framework could be used whenever additional market data was required. Establishing this reusable extraction layer became the foundation for testing continuity within the broader ML architecture.

Append and Integration Process

After extracting the new market data, the next step was integrating it into the existing workflow. This involved merging the newly collected data with the historical dataset while preserving the same schema structure and operational format used by the original workflow.

The focus during this stage was not on changing the architecture but on maintaining compatibility with it. The appended dataset was prepared using the same structural rules as the original data so that it could replace the existing raw-input layer without creating downstream inconsistencies. This process demonstrated that future data could be integrated into the workflow while preserving the conditions required for continuity.

Downstream Workflow Validation

Once the updated dataset was integrated, the next step was validating the downstream components of the workflow. Since the ML architecture operates through interconnected views, models, predictions, and analytical layers, the success of reusable continuity could not be judged solely by whether the data was appended successfully.

The workflow was therefore evaluated across multiple stages. Existing views continued operating correctly with the updated dataset. The machine learning models and prediction workflows remained compatible with the new data structure. The downstream analytical layers also continued functioning without requiring modification. Most importantly, these components survived through the existing architecture rather than through rebuilding or redesigning the workflow.

This became one of the strongest indicators that reusable continuity had been successfully achieved within the ML system.

Key Operational Understanding

The most important understanding from this stage was that reusable continuity is demonstrated through the survival of downstream dependencies rather than through data integration alone. Appending new data is only the first step. The real test is whether the architecture continues functioning after that data has entered the system.

Through this process, I observed that the ML workflow behaves as a highly centralized and dependency-connected architecture. The views, models, predictions, and analytical outputs remain closely linked through the same underlying structure. Because of this, continuity can survive successfully when schema compatibility is preserved. At the same time, the architecture remains sensitive to structural inconsistencies because a problem introduced at the input stage can affect multiple downstream components.

This understanding provided practical evidence that reusable continuity within the ML workflow depends on preserving both schema continuity and downstream compatibility.

7.4 Key Findings

The testing process demonstrated that future market data could be successfully extracted, integrated, and appended into the existing ML workflow using a reusable operational framework. The updated dataset remained compatible with the original schema structure, allowing new information to enter the workflow without requiring architectural changes.

The analysis further showed that the existing views, machine learning models, prediction workflows, and downstream analytical layers continued functioning after the updated data was introduced. These components remained operational through the existing architecture, demonstrating that continuity could be preserved without rebuilding the workflow.

Another important finding was that the ML workflow relies heavily on centralized dependencies. Because the architecture is highly interconnected, schema continuity and input-level compatibility play a critical role in protecting downstream functionality.

Overall, the testing provided practical evidence that the ML workflow can function as a reusable analytical system when future data is introduced through a structurally compatible process.

7.5 Interpretation

This stage transformed the idea of reusable continuity from a theoretical possibility into a practical observation. The earlier stages of the project suggested that continuity should survive if the architecture and schema were preserved, but this stage provided an opportunity to test that assumption within the workflow itself.

The results demonstrated that the ML workflow is capable of supporting future data continuity without requiring the existing models, views, prediction structures, or analytical layers to be rebuilt. The success of the workflow was not determined by the specific market data that entered the system, but by the ability of the architecture to maintain compatibility with that data. This reinforced the understanding developed during the previous stage that continuity depends more on preserving structure than on preserving the data itself.

At the same time, the testing highlighted the nature of the ML architecture as a centralized dependency-connected system. Its strength comes from the ability of multiple components to operate together through a common structure, but this also makes schema continuity particularly important. Overall, this stage demonstrated that reusable continuity within the ML workflow is achievable when structural compatibility and downstream continuity are preserved throughout the existing architecture.

8. Major Question 4 — Testing Reusable Continuity in the NLP Workflow

8.1 The Question

After testing reusable continuity within the ML workflow, the final major question focused on determining whether a similar form of continuity could be achieved within the NLP workflow. Unlike the ML system, which operates through a highly connected architecture inside BigQuery, the NLP workflow depends on external Python processing before the generated outputs are integrated back into SQL for further analysis. This creates a different operational structure and raises a different question. The objective of this stage was therefore to determine whether future news and earnings data could be introduced into the existing NLP workflow while preserving the continuity of the signal-generation and downstream analytical processes without rebuilding the overall architecture.

8.2 Why This Stage Was Needed

The successful continuity testing of the ML workflow provided evidence that a centralized architecture can remain reusable when future data is introduced through a structurally compatible process. However, the NLP workflow follows a different operational model. Rather than relying primarily on interconnected views and models, it depends on multiple extraction, processing, and signal-generation stages before the outputs become available for downstream analysis.

Because of this difference, the conclusions drawn from the ML workflow could not automatically be applied to the NLP workflow. The reusable continuity of the NLP system needed to be tested independently. Future news articles, earnings discussions, and textual narratives are expected to evolve continuously over time, making the workflow dependent on repeatable extraction and signal-generation processes. This stage therefore focused on understanding whether the existing NLP architecture could continue functioning when future text data was introduced through the same operational framework.

8.3 Analysis and Understanding

Reusable Data Extraction Framework

The first step involved creating reusable processes for collecting future textual data. For news data, a Google News RSS-based framework was used to support repeatable extraction of future articles. For earnings-related data, the existing extraction process was reviewed to determine how future earnings information could continue entering the workflow.

Rather than treating data collection as a one-time activity, the objective was to create repeatable extraction processes that could be executed whenever new data became available. This established the foundation for continuity because future news and earnings information could enter the workflow through the same operational structure used by the original project.

Signal Generation and Regeneration Process

After the new data was collected, the next step was regenerating the NLP outputs. The workflow relied on multiple NLP processes, including sentiment analysis, entity extraction, phrase extraction, and theme-related analysis. These outputs were generated through Python-based processing before being integrated into BigQuery SQL.

The focus during this stage was not on changing the signal-generation logic but on determining whether the existing processes could continue producing compatible outputs when future data was introduced. The regenerated sentiment, entity, and phrase outputs followed the same structural format as the original workflow, allowing them to remain compatible with the downstream analytical layers. This demonstrated that continuity could be maintained not only at the data level but also at the signal-generation level.

Downstream Workflow Validation

Once the regenerated outputs were integrated into the workflow, the next step was validating the downstream analytical processes. The objective was to determine whether the existing SQL-based analysis layers could continue functioning when supplied with newly generated NLP outputs.

The validation process showed that the downstream workflow remained operational when the regenerated signals followed the expected structural format. Existing analytical views and reporting layers continued functioning without requiring major redesign. Similar to the ML workflow, the continuity test was therefore evaluated not by whether the new data could be collected, but by whether the existing analytical architecture continued operating after the updated outputs entered the system.

This became one of the strongest indicators that reusable continuity had been achieved within the NLP workflow.

Key Operational Understanding

The most important understanding from this stage was that continuity within the NLP workflow operates differently from continuity within the ML workflow. The ML architecture survives through a centralized dependency-connected structure, whereas the NLP workflow survives through a modular signal-generation structure.

Because the NLP workflow contains multiple independent processing stages, continuity depends on preserving compatibility across these stages rather than maintaining a single centralized architecture. The workflow can therefore remain reusable even when individual extraction processes are executed separately, provided that the generated outputs continue to follow the structural requirements expected by the downstream analytical layers.

At the same time, the analysis also revealed that not every component of the workflow achieved the same level of reusability. While sentiment, entity, and phrase-related outputs could be regenerated through repeatable processes, theme interpretation and theme grouping continued to require analytical judgment. This indicated that the workflow achieved strong operational continuity, but some analytical interpretation layers still retained a manual component.

8.4 Key Findings

The testing process demonstrated that future news and earnings data could be collected through reusable extraction frameworks and integrated into the existing NLP workflow. The regenerated outputs remained structurally compatible with the original workflow, allowing new information to enter the system without requiring major architectural changes.

The analysis further showed that sentiment signals, entity outputs, phrase outputs, and other NLP-generated features could be regenerated using the existing processing logic and supplied to the downstream analytical layers. The existing workflow remained operational without requiring the overall analytical structure to be rebuilt.

Another important finding was that the NLP workflow achieves continuity through modular processes rather than through a centralized dependency structure. As long as the extraction, processing, and output-generation stages remain structurally compatible, the workflow can continue supporting future data integration.

At the same time, theme interpretation and theme grouping remained partially dependent on analytical judgment. While the underlying signals could be regenerated successfully, certain higher-level interpretation activities still required manual involvement.

Overall, the testing provided practical evidence that the NLP workflow can function as a reusable analytical system, while also highlighting areas where complete automation has not yet been achieved.

8.5 Interpretation

This stage expanded the understanding of reusable continuity beyond the centralized architecture observed in the ML workflow. While the ML system demonstrated how continuity can survive through interconnected dependencies, the NLP workflow demonstrated that continuity can also survive through modular operational processes.

One of the most important observations was that future news narratives, market discussions, and earnings themes do not need to remain the same for continuity to survive. The workflow remains reusable because the extraction, signal-generation, and analytical processes continue operating through a consistent structure. This reinforces one of the central findings of the project: continuity depends more on preserving operational structure than on preserving the underlying data itself.

The results also showed that reusable systems do not always need to follow a single architectural model. The ML workflow and NLP workflow achieve continuity through different mechanisms, yet both are capable of supporting future data integration without requiring the overall architecture to be rebuilt. However, the NLP workflow also highlighted an important distinction between operational continuity and analytical interpretation. While much of the workflow proved reusable, some interpretation layers—particularly theme-related analysis—continued to require human judgment.

Overall, this stage demonstrated that reusable continuity within the NLP workflow is largely achievable through modular and repeatable processes, while also identifying the areas where manual analytical involvement still remains. This provided a more complete understanding of both the strengths and limitations of reusable NLP systems.

9. Overall Interpretation of the Project

The project began with a relatively simple question: could the existing ML and NLP workflows continue functioning when future data became available, or would the workflows need to be rebuilt each time new information entered the system? Initially, the project appeared to be focused on future data integration. However, as the analysis progressed, it gradually became clear that the real challenge was not the future data itself, but understanding the conditions that allow analytical systems to remain reusable over time.

One of the most important realizations from the project was that reusable continuity depends more on preserving structure than on preserving data. Future market data, news narratives, and earnings discussions are naturally expected to change. The objective of a reusable system is therefore not to preserve the same information indefinitely, but to preserve the structural conditions that allow new information to move through the existing workflow. This shifted the focus of the project away from the data itself and toward schema continuity, validation processes, and reusable ingestion frameworks.

The project also revealed that reusable continuity can be achieved through different architectural approaches. The ML workflow operates as a centralized dependency-connected system where views, models, predictions, and analytical layers remain closely linked through a common structure. The NLP workflow, on the other hand, operates through a modular signal-generation architecture where extraction, processing, and analytical stages function as separate but connected components. Although these workflows differ significantly in their design, both demonstrated the ability to support future data continuity when their operational requirements were preserved.

Another important understanding was that reusability should not be viewed as complete automation. Throughout the project, many components of both workflows proved capable of supporting future data integration through repeatable processes. At the same time, some analytical activities, particularly within theme-related NLP analysis, continued to require human judgment and interpretation. This highlighted an important distinction between operational continuity and analytical interpretation. A workflow can remain reusable even when certain analytical decisions continue to involve manual involvement.

The project further demonstrated that continuity can only be evaluated through the survival of downstream components. Successfully appending new data is not sufficient evidence of reusability. The stronger test is whether the views, models, prediction structures, signal-generation processes, and analytical layers continue functioning after new data enters the workflow. The successful continuity testing of both the ML and NLP systems showed that the existing architectures could support future data integration without requiring the overall workflows to be rebuilt.

Perhaps the most significant outcome of the project is that it changed the nature of the original question. What began as an attempt to understand whether future data could be added to existing workflows gradually evolved into an exercise in understanding how analytical systems themselves function. The project moved beyond data continuity and became an exploration of architecture, dependencies, validation, operational design, and long-term reusability.

Overall, the project suggests that reusable analytical systems are sustained not by preserving the same data, but by preserving the structural and operational conditions that allow future data to be integrated consistently. The findings demonstrate that both the ML and NLP workflows can function as reusable analytical systems, while also highlighting the importance of schema continuity, validation, and operational discipline in maintaining that reusability over time. This broader understanding ultimately became the most valuable outcome of the project.

10. Key Learnings from the Project

One of the most important learnings from this project was that reusability should not be viewed as complete automation. A workflow can remain reusable even when certain analytical activities continue to require human judgment. What matters is whether the core operational structure can continue functioning when future data becomes available.

The project also demonstrated that structure is often more important than data when building reusable analytical systems. Market behaviour, news narratives, and earnings discussions will naturally change over time. However, the workflows can continue operating successfully if future data follows the structural requirements expected by the system. This highlighted the importance of schema continuity as a foundation for long-term reusability.

Another important learning was that validation plays a critical role in protecting operational continuity. Validation is not simply a quality-control step. It helps ensure that future data remains compatible with the existing workflow before it reaches downstream models, views, signals, and analytical layers.

The project further showed that different analytical systems can achieve continuity through different architectural approaches. The ML workflow demonstrated a centralized dependency-connected form of continuity, while the NLP workflow demonstrated a modular signal-generation form of continuity. Despite these differences, both systems were capable of supporting future data integration when their structural requirements were preserved.

Finally, the project reinforced the importance of thinking beyond individual analyses and toward systems. What began as an attempt to understand future data continuity gradually became an exercise in understanding how reusable analytical systems are designed, maintained, and sustained over time. This broader perspective became one of the most valuable learnings from the project.

11. Limitations of the Project

Although the project successfully demonstrated reusable continuity across both the ML and NLP workflows, certain limitations remained within the overall architecture and operational process.

One limitation emerged within the NLP workflow, particularly in theme-related analysis. While sentiment, entity, and phrase-based outputs could be regenerated through repeatable processes, theme grouping and theme interpretation continued to require analytical judgment. As a result, this part of the workflow remained only partially reusable and still depended on manual involvement.

Another limitation relates to the earnings-data workflow. Compared with the news extraction process, earnings-related data remains more dependent on the availability, structure, and accessibility of external sources. This can affect the consistency and scalability of future earnings-data continuity.

The project also remains dependent on schema stability. Although reusable ingestion frameworks and validation processes were developed to protect continuity, significant structural changes in future datasets could still require adjustments to parts of the workflow. Reusability therefore depends on maintaining compatibility between future data and the existing architecture.

In addition, certain operational activities continue to require manual oversight. Data validation, continuity checks, and output verification remain important to ensure that the workflows continue functioning as expected when future data is introduced. While these activities are considerably smaller than rebuilding the workflow itself, they have not been fully eliminated.

Overall, the project demonstrated strong reusable continuity, but it also highlighted that reusable systems still depend on operational discipline, schema stability, and selective human involvement in specific analytical areas.

Table 2: Limitations Summary

Limitation	**Impact**
Theme interpretation and grouping	Requires continued analytical judgment and manual involvement
Earnings-data dependency	Future continuity depends on external data availability and consistency
Schema-change risk	Significant structural changes may require workflow adjustments
Manual validation requirements	Periodic oversight remains necessary to maintain continuity

12. Future Improvements

While the project successfully demonstrated reusable continuity across both the ML and NLP workflows, several opportunities exist to further strengthen the architecture and reduce remaining operational dependencies.

One area for improvement relates to theme-related analysis within the NLP workflow. Although sentiment, entity, and phrase-based outputs could be regenerated through repeatable processes, theme grouping and theme interpretation still relied on analytical judgment. Future work could focus on developing more structured approaches for automated theme classification and grouping, allowing a larger portion of the workflow to operate through repeatable processes.

A second improvement area involves the earnings-data workflow. Enhancing the extraction and preprocessing framework could improve the consistency and scalability of future earnings-data integration. A more robust and standardized extraction process would further strengthen continuity across future earnings-related analyses.

The project could also benefit from additional automation within the ingestion and validation layers. While reusable frameworks were developed for future data integration, further automation of compatibility checks and validation processes could reduce the amount of manual oversight required during workflow updates.

Another potential improvement involves strengthening resilience against future schema changes. More advanced validation rules and monitoring mechanisms could help identify structural issues earlier and improve the ability of the workflows to adapt to changing data environments.

Finally, future versions of the project could explore more real-time or near real-time data integration approaches. While the current architecture successfully supports reusable continuity through periodic updates, extending the framework toward more continuous data ingestion could further improve scalability and operational efficiency.

Overall, these improvements build upon the existing architecture rather than replacing it. The project has already demonstrated that reusable continuity is achievable, and future enhancements would primarily focus on increasing automation, scalability, and operational robustness.

Table 3: Future Improvements Summary

Area	**Possible Improvement**
Theme Analysis	More structured and automated theme grouping and classification
Earnings Workflow	Improved extraction and preprocessing framework
Validation Layer	Greater automation of compatibility and quality checks
Schema Monitoring	Earlier detection of structural changes and inconsistencies
Data Continuity	Exploration of real-time or near real-time update processes

13. Final Answer to the Problem Statement

The central objective of this project was to determine whether the Machine Learning and Natural Language Processing workflows that had been developed in the earlier projects could function as reusable analytical systems rather than remaining one-time analytical projects. More specifically, the project sought to understand whether future market data, news data, and earnings data could be introduced into the existing workflows while preserving the operational continuity of the underlying architecture.

Based on the analysis performed throughout the project, the answer is largely yes.

The project demonstrated that both the ML and NLP workflows are capable of supporting future data continuity without requiring the overall analytical architecture to be rebuilt. Through the development of reusable ingestion processes, schema validation approaches, and continuity testing frameworks, future data was successfully integrated into the existing workflows while preserving compatibility with downstream components.

Within the ML workflow, future market data was successfully extracted, appended, and integrated into the existing architecture. The continuity of views, models, prediction structures, and downstream analytical layers was preserved through the existing workflow. This demonstrated that the ML system could continue operating through the same architecture when future data was introduced in a structurally compatible manner.

Within the NLP workflow, future news and earnings data could be collected through repeatable extraction processes and regenerated into compatible analytical outputs. Sentiment, entity, and phrase-related signals continued to support downstream analytical processes without requiring the workflow to be rebuilt. This demonstrated that the NLP system could also support reusable continuity, although it achieved that continuity through a more modular architecture than the ML workflow.

At the same time, the project also showed that reusable continuity should not be interpreted as complete automation. Certain analytical activities, particularly within theme-related NLP analysis, continued to require human judgment and interpretation. As a result, the project demonstrated strong reusable continuity, but not complete analytical automation. This distinction became one of the most important findings of the entire project.

Perhaps the most significant outcome of the project was that it changed the nature of the original question. What began as an attempt to determine whether future data could be integrated into existing workflows gradually became an exercise in understanding how analytical systems maintain continuity over time. The project ultimately showed that long-term reusability is less about preserving data and more about preserving the structures, validation mechanisms, and operational processes that allow future data to move through an existing architecture. In this sense, the project evolved beyond a continuity-testing exercise and became an exploration of how reusable analytical systems can be sustained over time.

Overall, the project successfully demonstrated that both the ML and NLP workflows can function as reusable analytical systems when future data is introduced through compatible and repeatable processes. While certain limitations remain, particularly within higher-level analytical interpretation, the findings provide strong evidence that the existing architectures can support future continuity without requiring the entire workflow to be rebuilt. The project therefore achieved its primary objective while also providing a broader understanding of the principles that support long-term reusability in analytical systems.

14. Appendix / Supporting Assets

The main report presented here is based on a detailed working document developed during the actual execution of the project. This underlying document captures the step-by-step analysis, observations, testing activities, and interpretations that were carried out while evaluating the reusable continuity of both the ML and NLP workflows.

This working document is included in the project repository as the **Reusable Systems Working Analysis** file. It reflects the complete analytical process followed throughout the project, including the progression from understanding the existing architecture and schema requirements to testing reusable continuity across the ML and NLP systems.

Supporting assets, including the processed datasets, validation outputs, reusable SQL components, and reusable Python frameworks developed during the project, are also included in the repository. These files provide additional context on how future data was extracted, integrated, validated, and evaluated within the existing analytical architecture.

Overall, these supporting materials provide a complete view of both the analytical process and the operational implementation of the project, helping to explain how the conclusions presented in this report were developed and validated.